

unlike other mobile devices that users can turn on manually when needed. To preserve battery life, wearables must run at an extremely low power. This demand creates unique requirements for the controller and firmware algorithms.

A way to tackle this issue is to offload the processing power to the cloud using a mobile or web application. In the case of a mobile application, which is more common, Bluetooth is a great way to ensure connectivity between the wearable device and your smartphone.

In terms of hardware, there are many controllers that work on extremely low power. A good example would be the PSoC family of programmable microcontrollers. These are recommended because they utilise the strength of the ARM architecture by combining an ARM Cortex-M core with advanced analogue and programmable digital capability on a single chip.

Qualcomm, in July 2022, unveiled their latest chips for wearables—the Snapdragon W5+ Gen 1 and the Snapdragon W5 Gen 1. According to Qualcomm, these chips have been designed to advance ultra-low power and breakthrough



Snapdragon W5+ and W5 Gen 1 wearable platforms launched in July 2022 (Credit: Qualcomm)

performance for next-generation connected wearables with a focus on extended battery life.

Co-processor and multicore processors. To offload the processing of sensor data from the main CPU, certain cutting-edge devices have a separate co-processor. This is necessary when the device needs to examine a lot of sensor data simultaneously in real time while demanding constant CPU attention.

Not just the sensor data, many wearable devices actually have co-processors or DSPs for specialised applications like AI and ML, so that these things can also be done on the edge.

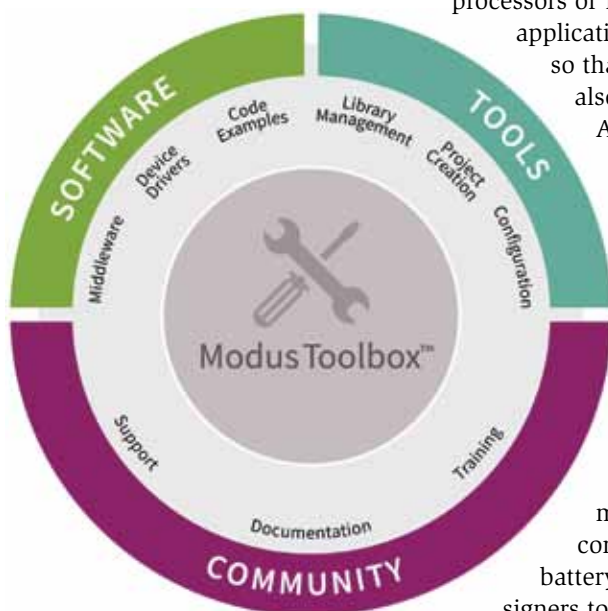
As mentioned above, the Snapdragon W5+ Gen 1 and the Snapdragon W5 Gen 1 chips have an ML co-processor (ARM Ethos-U55).

Similarly, Renesas' DA1469x product family contains three new cores that provide more resources, range, computing power, and battery life—allowing designers to push the boundaries of applications like weara-

bles. In June 2022, Renesas also launched the SmartBond DA1470x family of SoCs. It is a multi-core system with CM33F as the main application core and CM0+ as the sensor node controller. They claim that this is the world's first SoC to offer a very high level of low power integration and is best suited for wearable medical devices.

Security and development environment. Modern wearables—especially medical monitoring devices—deal with a lot of sensitive data, like biometric information, and then transmit this wirelessly. To protect user data, these wireless connections must be private and secure. A strong yet simple-to-implement security framework is a very crucial feature in microcontrollers.

Furthermore, a comprehensive development ecosystem makes it easier for developers to integrate different operating systems and software for a nicer, smartphone-like experience for wearable devices. For example, NXP's i.MX RT microcontrollers provide built-in security features, including secure boot, one-of-a-kind key storage, and hardware acceleration of symmetric and cryptographic algorithms to help protect smart-



Features provided by ModusToolbox (Credit: Infineon)